

Name: _____

Roll Number: _____

Quiz-1**Max. Time: 20 mins****Max. Marks: 15**

Note: Solve all parts. Limit your written responses to the provided space.

Q.1. [5] Choose by putting a check mark on the most appropriate option. Note: No cutting/overwriting allowed.

i. Time complexity of DFID algorithm can be characterized as:

☒ (A) $O(d^b)$ (B) $O(b^d)$ (C) $O(bd)$ (D) None of the given

ii. For the 8-puzzle problem, Euclidean distance is a more informed heuristic than the hamming distance.

☒ (A) True (B) Falseiii. If $g(n)$ varies from node to node in the state-space graph of a problem, then the BFS algorithm:☒ (A) may return a sub-optimal path (B) returns the same path as Dijkstra's algorithm (C) must use an admissible heuristic (D) None of the given choices

iv. _____ is the algorithm of choice while searching for a goal in uninformed way.

(A) Best first search (B) BFS ☒ (C) DFID (D) A*

v. A* algorithm, which uses a heuristic that doesn't guarantee admissibility, still returns an optimal path.

(A) True ☒ (B) False**Q.2. [3]**

a) Prove that a monotone heuristic is always admissible.

A simple argument can show that any monotonic heuristic is admissible. This argument considers any path in the space as a sequence of states $s_1, s_2, \dots, s_g$, where s_1 is the start state and s_g is the goal. For the sequence of moves in this arbitrarily selected path:

s_1 to s_2	$h(s_1) - h(s_2) \leq \text{cost}(s_1, s_2)$	by monotone property
s_2 to s_3	$h(s_2) - h(s_3) \leq \text{cost}(s_2, s_3)$	by monotone property
s_3 to s_4	$h(s_3) - h(s_4) \leq \text{cost}(s_3, s_4)$	by monotone property
.	.	by monotone property
.	.	by monotone property
s_{g-1} to s_g	$h(s_{g-1}) - h(s_g) \leq \text{cost}(s_{g-1}, s_g)$	by monotone property

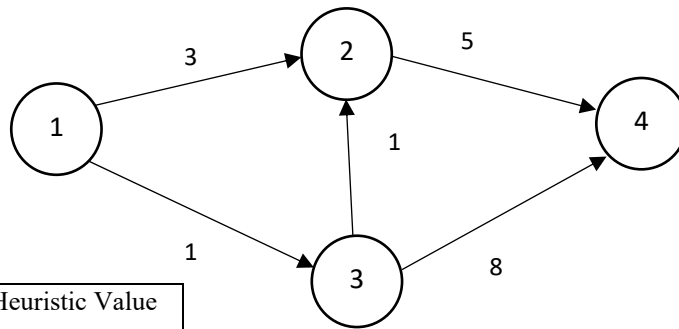
Summing each column and using the monotone property of $h(s_g) = 0$:

$$\text{path } s_1 \text{ to } s_g \quad h(s_1) \leq \text{cost}(s_1, s_g)$$

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Q.3. [7] For the following directed graph, run the A* algorithm using the given heuristic function and state whether the heuristic is consistent or not. Show the contents of the OPEN and CLOSED lists, as used with the best first search algorithm, using the hyphenated notation, e.g., for state 1, use 1-5, where '1' is the state and '5' is the heuristic estimate. Note that state '4' is the goal state.



State	Heuristic Value
1	5
2	1
3	6
4	0

Solution: The given heuristic is not monotone because for states '1' and '2', the following inequality doesn't hold.

$$h(1) - h(2) = 5 - 1 = 4 \not\leq \text{cost}(1,2) = 3$$

The result of running A* algorithm is shown in the table below.

<u>OPEN</u>	<u>CLOSED</u>
[1-5]	[]
[2-4,3-7]	[1-5]
[3-7,4-8]	[1-5,2-4]
[2-3,4-8]	[1-5,3-7]
[4-8]	[1-5,3-7,2-3]
Goal Found!	

Solution path: 1->3->2->4.